

✅ DAY 2 — PDF INGESTION + PARSING (FINAL, USING YOUR STACK ONLY)

✅ 🎯 WHERE WE ARE

Infra ✅

API ✅

Agents ✅

Parser ❌

✅ 🎯 TODAY GOAL

- integrate **Unstructured OR LlamaParse**
 - extract **structured elements (not raw text only)**
 - prepare for tables + text pipeline
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✅ 📁 FILES

app/ingestion/

└─ pipeline.py

app/parser/

└─ pdf_parser.py

✅ STEP 1 — INSTALL (CHOOSE ONE)

✅ Option 1 (OSS)

```
pip install "unstructured[pdf]"
```

✅ Option 2 (PRO)

```
pip install llama-index llama-parse
```

```
``
```

✅ STEP 2 — PDF PARSER (UNSTRUCTURED VERSION)

app/parser/pdf_parser.py

```
from unstructured.partition.pdf import partition_pdf
```

```
def parse_pdf(file_path):  
    elements = partition_pdf(filename=file_path)  
  
    parsed = []  
  
    for el in elements:  
        text = getattr(el, "text", "").strip()  
        if not text or text.isdigit():  
            continue  
        parsed.append({  
            "type": type(el).__name__,  
            "text": text,  
            #"metadata": el.metadata  
        })  
    return parsed
```

STEP 3 — PIPELINE

app/ingestion/pipeline.py

```
from app.parser.pdf_parser import parse_pdf
```

```
def process_pdf(file_path):  
    elements = parse_pdf(file_path)  
  
    return {  
        "elements": elements  
    }
```

✅ STEP 4 — TEST

 **tests/test_ingestion.py**

```
from app.ingestion.pipeline import process_pdf
```

```
result = process_pdf("data/sample.pdf")
```

```
for e in result["elements"][:10]:
```

```
    print(e)
```

✅✅ EXPECTED OUTPUT

You will see:

```
{'type': 'Title', 'text': 'TPDS TEST REPORT...'}
```

```
{'type': 'NarrativeText', 'text': 'Objective: ...'}
```

```
{'type': 'Table', 'text': 'Notch Effort Travel...'}
```

✅✅ IMPORTANT RULES

👉 DO:

extract structured elements ✅

keep type info ✅

👉 DO NOT:

use PyMuPDF ❌

flatten everything into one string ❌

hardcode tables ❌

✅🚀 STATE AFTER DAY 2

Infra ✅

API ✅

Agents 

Structured parsing  (text + table tags)

Table extraction  (not yet structured)

Schema 

Analytics 

  SOON **NEXT (DAY 3)**

 Build:

table extraction → clean rows/columns (DocLing / Unstructured tables)

 Say **Day 3** when ready 